LLM6G: A Shift-Aware Forecast-to-Control Pipeline for 6G Mobile Network Traffic

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Abstract*—Mobile network traffic evolves through regimes separated by distribution shifts. Controlling the network over the next stable interval requires predicting where the next break occurs and bounding the upper traffic path up to that break. We propose LLM6G, a general shift-aware forecast-to-control pipeline where any probabilistic forecaster plugs into a shared interface: forecast the next 8 hours, detect the first break on the median path, compute a safe ceiling from the upper quantile up to that break, and refresh after the predicted break. We instantiate the pipeline with five forecasters on the public TIM Milan benchmark (200 trainable cells, native 10-minute cadence, 144-step daily-cycle context). Coverage and relative sharpness are competing objectives: no model dominates both. The Temporal Fusion Transformer (TFT) achieves the best trained operating point (RMSE 5.658, q95 coverage 0.924, relative sharpness 0.153: 15.3% excess above realized demand). Chronos-2, with no domain training, achieves the tightest safe ceiling in the benchmark (relative sharpness 0.082: 8.2% excess) at competitive CP timing ($\mathrm{MAE}_{\mathrm{CP}}$ 7.415) — demonstrating that LLM-based foundation models can plug directly into 6G-oriented control loops. We document three implementation decisions — channel-selective normalization, continuous likelihood matching, and quantile-based checkpoint monitoring — that determine whether the pipeline meets coverage targets or fails silently.

***Index Terms*—traffic forecasting, change-point detection, probabilistic forecasting, network control, TIM Milan, Chronos-2, Temporal Fusion Transformer, 6G**

I. Introduction

Traffic demand in mobile networks evolves through regimes. Daily cycles, weekends, event surges, and broader demand changes shift both the level and the shape of the traffic distribution. These shifts matter directly for control. A policy built around the current load level alone can keep the network at an unsafe or overly conservative operating point when the next regime change arrives. The relevant control object is not the next-step forecast. It is the next stable operating interval and the safe ceiling over that interval.

Most existing work addresses this problem in parts. Traffic forecasting papers optimize accuracy metrics but stop before the control decision [1], [2]. Network control papers set sleeping or provisioning policies around predicted load levels without explicitly modeling the next distribution break [3]–[5]. What is missing is an end-to-end shift-aware pipeline that makes the next break a first-class control object, evaluated on public data under shared metrics.

This paper fills that gap. We present LLM6G: a general forecast-to-control pipeline where any probabilistic forecaster plugs into a shared interface. The pipeline forecasts a probabilistic traffic path, detects the first break on the median forecast, computes a safe ceiling from the upper quantile up to that break, and refreshes after the predicted break. The name reflects the two components: the LLM (Chronos-2, a foundation model for time series) and the 6G control objective. We use "LLM" in the broad sense of large pre-trained generative models for sequential data, consistent with the emerging literature on LLMs for time series [6], [7].

We benchmark LLM6G on the public TIM Milan dataset [8] at its native 10-minute cadence. The dataset remains active in the public forecasting literature [9], [10] and provides a clean non-augmented telecom traffic proxy at city scale. A neutral 200-cell trainable subset enables direct comparison between trained and zero-shot forecasters.

Five forecasters are evaluated inside the pipeline: the Temporal Fusion Transformer (TFT) [11], LSTM, DeepAR [12], Chronos-2 [7], and Seasonal Naive. Coverage and sharpness of the safe ceiling are the primary control metrics. Neither dominates: a tighter ceiling reduces over-provisioning but risks coverage violations. TFT achieves the best trained operating point. Chronos-2 achieves the tightest ceiling across all five models with no domain training, supporting its role as a zero-shot foundation model for 6G control under recurring distribution shifts.

The paper makes four contributions:

- **LLM6G pipeline.** A general model-agnostic forecast-to-control framework: probabilistic forecast, first-break detection, pre-break safe ceiling, receding-horizon refresh. Any forecaster that produces $q_{0.5}$ and $q_{0.95}$ plugs in.
- **TFT as best trained model.** Channel-selective instance normalization enables TFT to generalize across distribution shifts. Without it, the validation-to-test RMSE gap exceeds 30%.
- **Chronos-2 in the control loop.** First evaluation of Chronos-2 in a shift-aware forecast-to-control pipeline. It achieves the tightest safe ceiling (relative sharpness 0.082: 8.2% excess above demand) with competitive CP timing and no domain training.
- **Public TIM Milan benchmark.** 200-cell trainable subset, 144-step daily-cycle context, native 10-minute cadence, 5-model evaluation under a shared canonical protocol.

II. Related Work

A. Telecom traffic prediction and public benchmarks

Public telecom traffic benchmarks remain limited. The TIM Milan release by Barlacchi *et al.* [8] is one of the most reusable public datasets for city-scale telecom activity. Recent work continues to use it to evaluate stacked recurrent models, transformers, and TimeGPT-style forecasters [9], [10]. Surveys of cellular traffic prediction confirm the same point: the forecasting literature is broad, but reproducible public benchmarking is thinner than proprietary operator evaluation [1], [2]. Several papers apply spatial-temporal deep models on grid-level data [13]–[15]. Our paper keeps the public-benchmark perspective and shifts from forecasting accuracy alone to a control loop centered on the next predicted regime change.

B. Forecasting for network control and energy-aware operation

Traffic prediction is a standard ingredient in energy-aware network control. Recent work has applied it to green cellular networking [16]–[18], base-station sleeping [3], [4], and digital-twin-assisted management [5]. These papers motivate the operational value of forecasting. They typically optimize around predicted load levels or sleeping decisions, however, rather than around the next predicted distribution break. Our pipeline sits one level above point prediction: it explicitly predicts the next regime boundary and sets a safe ceiling only over the resulting stationary interval.

C. Neural probabilistic forecasters, foundation models, and change detection

DeepAR [12] models per-series distributions autoregressively and extracts quantiles from Monte Carlo samples at inference. The Temporal Fusion Transformer [11] adds multi-head self-attention with variable selection networks and interpretable gating, achieving strong multi-horizon quantile forecasting; it has also been applied directly to network traffic prediction [15]. Both are trained baselines in this paper. Chronos [6] and Chronos-2 [7] bring the foundation-model paradigm to time series: pretrain on a large diverse corpus and deploy without target-specific retraining. Concurrent work applies time-series foundation models to ISP and optical network telemetry [2]; these focus on forecasting accuracy rather than control-oriented evaluation. For our setting, zero-shot deployment is operationally attractive because recurring traffic shifts can make periodic retraining costly.

Change-point detection provides a direct way to reason about regime boundaries. PELT [19] is an efficient exact change-point detector. CLASP [20] classifies sliding windows via k-nearest-neighbour accuracy, returns an explicit no-break decision when no score threshold is exceeded, and requires no assumption about the cost structure of change points. We select CLASP through a validation sweep and use it as the shared detector for all forecasters.

What is missing in the public telecom setting is an end-to-end benchmark connecting probabilistic forecasting, break detection, safe ceiling computation, and control-oriented evaluation. The contribution of this paper is that pipeline and its instantiation on a public benchmark. We do not propose a new forecasting architecture.

III. The LLM6G Forecast-to-Control Pipeline

A. Design philosophy

The pipeline is intentionally simple. Its value is not in the forecasting model — any calibrated forecaster plugs in. Its value is in making the next predicted distribution break a first-class control object. Existing control policies use forecasts to set a resource budget for the full horizon. This policy sets a tighter budget valid only over the next stable sub-interval and refreshes when that interval ends.

Five forecasters are benchmarked inside the same pipeline: TFT, LSTM, DeepAR, Chronos-2, and Seasonal Naive. They differ in architecture, training regime, and how they produce uncertainty estimates. All share the same output interface: a median path $q_{0.5}(1:H)$ and an upper path $q_{0.95}(1:H)$ over horizon H.

B. Pipeline stages

Figure 1 shows the full pipeline. Given a traffic context $x_{t-C+1:t}$ of length C:

Stage 1 — Probabilistic forecast. The forecaster produces two paths over the next H steps:

$$q_{0.5}(1:H) \quad \text{and} \quad q_{0.95}(1:H).$$

Stage 2 — First-break detection. A CLASP change-point detector [20] is applied to the median path $q_{0.5}$. It returns the first predicted break point τ_{pred}, or H if no break is detected.

Stage 3 — Safe ceiling. The pre-break safe ceiling is

$$c_{\text{safe}} = \max_{1 \le k \le \tau_{\text{pred}}} q_{0.95}(k).$$

This ceiling covers the predicted stationary interval using the upper quantile path up to the predicted break.

Stage 4 — Control and refresh. The network is provisioned up to c_{safe} until τ_{pred}. At that point, the pipeline refreshes: it collects new observations, updates the context, and runs stages 1–3 again. This yields a receding-horizon loop driven by predicted regime changes.

C. Why explicit break detection matters

Provisioning the network up to a ceiling computed from the full horizon overestimates the requirement over the early stationary part of that horizon. The ceiling should reflect only the traffic regime that is expected to be in place. By restricting the ceiling computation to $[1, \tau_{\text{pred}}]$, the pipeline produces a tighter and more accurate target as long as the break is predicted well.

This separates two objectives that are often conflated: forecast quality over the full horizon and control quality over the next stable interval. A model with good RMSE can still produce a poor control interval if its break point or upper quantile is misplaced.

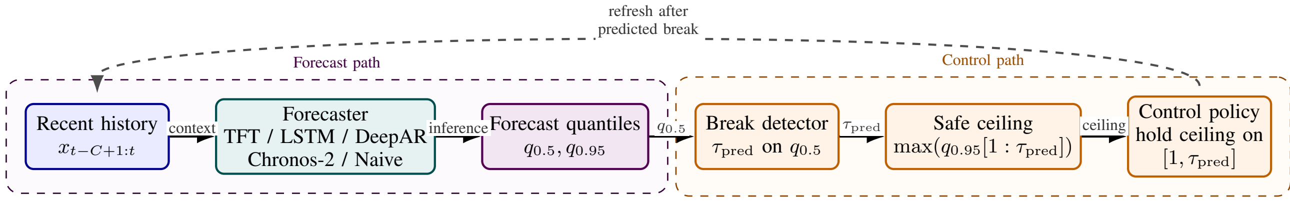

Fig. 1. The LLM6G forecast-to-control pipeline. Any forecaster that produces a median path $q_{0.5}$ and an upper path $q_{0.95}$ over horizon H plugs into the shared interface. The five forecasters evaluated in this paper — TFT, LSTM, DeepAR, Chronos-2, Seasonal Naive — are interchangeable blocks at the left. The downstream controller and evaluation logic are identical across all models.

IV. Problem Formulation

Let $x_{t-C+1:t}$ denote a context window of length C and let H denote the forecast horizon. A forecaster receives the context and produces two paths over the next H steps: the median path $q_{0.5}(1:H)$ and the upper path $q_{0.95}(1:H)$.

The control loop focuses on the first predicted regime change within the horizon. Let

$$\tau_{\text{pred}} = \mathrm{CP}(q_{0.5}(1:H)), \tag{1}$$

where $\mathrm{CP}(\cdot)$ returns the first detected change point on the median path, or H if no earlier change is found. The pre-break safe ceiling is

$$c_{\text{safe}} = \max_{1 \le k \le \tau_{\text{pred}}} q_{0.95}(k). \tag{2}$$

For evaluation, the same detector is applied to the realized future path to obtain τ_{true}. The pipeline is assessed with four quantities:

- *Change-point timing error*: $\mathrm{MAE}_{\mathrm{CP}} = |\tau_{\text{pred}} - \tau_{\text{true}}|$, averaged over windows.
- *Tolerance hit rate*: fraction of windows where $|\tau_{\text{pred}} - \tau_{\text{true}}| \le 3$ steps (30 minutes at 10-minute cadence).
- *Coverage rate*: fraction of realized pre-change points that stay below c_{safe}.
- *Relative sharpness*: average of $(c_{\text{safe}} - \max(\mathcal{I}_{\text{true}}))/\max(\mathcal{I}_{\text{true}})$ over windows, where $\mathcal{I}_{\text{true}}$ is the realized pre-change interval. Expresses excess ceiling as a fraction of realized demand.

Formally:

$$\mathrm{CoverageRate} = \frac{\sum_{x \in \mathcal{I}_{\text{true}}} \mathbf{1}[x \le c_{\text{safe}}]}{|\mathcal{I}_{\text{true}}|}, \tag{3}$$

$$\mathrm{RelSharpness} = \frac{c_{\text{safe}} - \max(\mathcal{I}_{\text{true}})}{\max(\mathcal{I}_{\text{true}})}. \tag{4}$$

Coverage and relative sharpness are competing objectives. A forecaster that always sets a wide ceiling achieves high coverage at the cost of over-provisioning. A forecaster that sets a tight ceiling reduces wasted capacity but risks ceiling violations. Small positive relative sharpness is ideal. Negative relative sharpness is a control failure: the ceiling already lies below the realized traffic peak before the break.

We report both metrics without collapsing them. The trade-off is a property of the forecaster, not an artifact of the evaluation.

TABLE I
TIM Milan benchmark configuration.

Quantity	Value
Complete grid cells	8,883
Trainable subset	200 cells
Total timestamps	8,928
Cadence	10 min
Date range	2013-11-01 to 2014-01-01
Context length C	144 steps (24 h)
Horizon H	48 steps (8 h)
Train / val / test rows	6,249 / 893 / 1,786
Test windows	7,400

V. Public TIM Milan Benchmark

The benchmark is built from the TIM Milan telecom activity dataset released by Barlacchi *et al.* [8]. Raw archives are accessed through the public O-RAN SC dataset mirror. Each record contains a Milan grid-cell identifier, a timestamp, and an `InternetTraffic` value. We use only this field.

The benchmark keeps the original 10-minute cadence. The harness concatenates daily archives, aligns the city-wide time grid, and retains only fully observed cells. No imputation is applied. The complete-cell benchmark contains 8,883 cells and 8,928 timestamps from 2013-11-01 through 2014-01-01.

For the experiments in this paper, a trainable subset of 200 cells is used. The subset is neutral: it keeps the first 200 complete cells by ascending grid-cell identifier. This rule avoids activity-based cherry-picking.

Context length. The context is set to $C = 144$ steps, which spans exactly one full daily cycle at 10-minute cadence. A 24-hour context window provides the prior day's complete traffic pattern as input. Shorter contexts miss the dominant periodicity; longer contexts risk including multiple regime changes in the conditioning window.

The benchmark represents public urban telecom activity and not operator base-station counters. All claims are at the level of grid-cell telecom proxies, not final base-station validation.

VI. Forecasters and Implementation

All forecasters share the same output interface: a median path and an upper path over horizon $H = 48$ steps. This section describes each model and documents the implementation decisions that had measurable impact on control performance.

A. Temporal Fusion Transformer

TFT [11] is the best-performing trained model in this benchmark. Its architecture combines variable selection networks (VSN) that learn which inputs are relevant at each step, an encoder-decoder LSTM (hidden size 128, 2 layers, dropout 0.1), interpretable multi-head self-attention (4 heads, shared value projection, causal mask), gated residual networks (GRN) for feature enrichment, and a direct quantile output head producing $q_{0.5}$ and $q_{0.95}$.

Time features — cyclic encodings of hour-of-day and day-of-week for the 10-minute cadence — are concatenated with the raw traffic value to form a multi-channel input tensor.

Channel-selective instance normalization. TFT receives a multi-channel input (raw value on channel 0; time features on channels 1+). Time features encode calendar position and must not be rescaled. We normalize only the raw-value channel:

$$\text{scale} = \text{mean}(|x_{1:C,0}|) + 1, \tag{5}$$

divide channel 0 by *scale* before encoding, and rescale the output by *scale* before computing the loss. The $+1$ prevents division by zero on constant or sparse series.

Without this fix, TFT memorizes absolute traffic levels instead of learning scale-invariant dynamics. The validation-to-test RMSE gap exceeds 30% without normalization, because the test period has different average traffic magnitudes than the validation period. With normalization, the gap closes to under 2%.

Training uses pinball loss at $q \in \{0.5, 0.95\}$. The checkpoint monitor is $\text{pinball}_{0.5} + \text{pinball}_{0.95}$ on the validation set.

B. Chronos-2 (LLM baseline, zero-shot)

Chronos-2 [7] is a foundation model for time-series forecasting based on quantized token autoregression. It is pretrained on a large diverse corpus of public time series and deployed without any Milan-specific fine-tuning. Quantile forecasts are extracted from sampled token paths.

In the LLM6G framing, Chronos-2 is the LLM component: a broadly trained model deployed directly into a 6G-oriented control loop. The key operational property is that it requires no retraining when the traffic distribution shifts — a practical advantage in networks that undergo recurring regime changes. We use the `amazon/chronos-2-small` checkpoint.

C. LSTM (trained baseline)

The LSTM forecaster uses a 2-layer LSTM (hidden size 128, dropout 0.1) followed by a direct quantile output head that produces all $H \times 2$ outputs in a single forward pass. Instance normalization is applied to the context:

$$\text{scale} = \text{mean}(|x_{1:C}|) + 1. \tag{6}$$

The context is divided by *scale* before encoding; outputs are rescaled accordingly. Training uses pinball loss.

D. DeepAR (cautionary baseline)

DeepAR [12] uses an autoregressive LSTM with per-series item embeddings (dimension 20), time features (hour-of-day, day-of-week, log-age), and a Gaussian parameter head. Quantile forecasts are extracted from 200 Monte Carlo samples at inference.

Likelihood choice. DeepAR supports multiple likelihoods, including Gaussian for real-valued data and Negative Binomial for count data. Milan traffic values are continuous grid-cell aggregates; we use Gaussian likelihood as appropriate. However, Gaussian confidence intervals prove too narrow for the heavy-tailed traffic distribution.

The consequence, however, is that Gaussian confidence intervals prove too narrow for the heavy-tailed traffic distribution: q95 coverage on the test set is 0.667, and relative sharpness is -0.041 (the ceiling falls 4.1% below the realized demand peak). A negative relative sharpness means the safe ceiling falls below the realized traffic peak before the break — the pipeline under-provisions rather than over-provisions. This is a control failure. We include DeepAR as a documented case of correct architecture with a likelihood mismatch for this data type.

Checkpoint monitoring. Initial experiments used Gaussian NLL as the early-stopping monitor. NLL improved steadily while q95 coverage stagnated at 0.64: the best-NLL checkpoint does not correspond to the best quantile coverage. Switching the monitor to $\text{pinball}_{0.5} + \text{pinball}_{0.95}$ selects checkpoints that directly maximize quantile quality.

E. Seasonal Naive (lower baseline)

The Seasonal Naive forecaster requires no training. The median forecast is the lag-144 value (yesterday-at-the-same-time). The upper-path forecast is the median plus the 95th percentile of historical differences $\{x_t - x_{t-144}\}$ computed over the context window. This provides a simple operational heuristic and an interpretable lower bound.

F. CLASP change-point detector

CLASP [20] detects the first break point non-parametrically. For each candidate τ, it classifies sliding windows as pre-change or post-change using k-nearest-neighbour accuracy on z-normalized window features. It returns the τ with the maximum classification score if that score exceeds a threshold; otherwise it returns "no break." This explicit no-break decision is important: a forecaster that never predicts a break applies the ceiling to the full horizon, which is correct when the traffic regime is stable.

CLASP is selected over PELT because it produces an explicit no-break output via score thresholding and requires no assumption about the cost structure of change points. Parameters are tuned via a validation sweep over 36 combinations (min_size $\in \{8, 10, 12\}$, period_length $\in \{8, 16, \text{auto}\}$, score_threshold $\in \{0.5, 0.6, 0.7, 0.75\}$). The best configuration is shared across all models: min_size $= 8$, period_length $= 16$, score_threshold $= 0.5$. Note that break F1 saturates at

TABLE II
MODEL ROLES AND TRAINING REGIMES.

Model	Training	Role
TFT	supervised	best trained
LSTM	supervised	trained baseline
DeepAR	supervised	cautionary
Chronos-2	zero-shot	LLM baseline
Seasonal Naive	none	lower bound

1.0 on training data for all configurations; selection is therefore based on τ-MAE rather than F1.

VII. EXPERIMENTAL SETUP

All experiments use the trainable TIM Milan subset: 200 cells, native 10-minute cadence, context length $C = 144$ steps (24 hours), horizon $H = 48$ steps (8 hours). The dataset is split chronologically in 70/10/20 proportion, yielding 6,249 train rows, 893 validation rows, and 1,786 test rows. All models share the same split. Evaluation uses non-overlapping horizon-step windows (step $= 48$) anchored at split start, yielding 7,400 test windows (200 series $\times$ 37 windows) and 3,600 validation windows. All models are evaluated on identical windows, removing sampling variance when comparing trained and zero-shot forecasters.

A. Trained model hyperparameters

TFT, LSTM, and DeepAR share the recurrent backbone: two layers, hidden size 128, dropout 0.1. TFT adds 4-head interpretable self-attention with a shared value projection and a causal mask. All three models output quantiles at $\{0.5, 0.95\}$.

Training uses the Adam optimizer with pinball loss at $q \in \{0.5, 0.95\}$. The early-stopping monitor is

$$\mathcal{L}_{\text{monitor}} = \text{pinball}_{0.5} + \text{pinball}_{0.95}$$

evaluated on the validation set. Training runs for a maximum of $\sim$95,000 iterations with patience $\sim$9,500 iterations, evaluated every $\sim$4,750 iterations. The checkpoint with the lowest monitor value is retained.

B. Chronos-2 setup

Chronos-2 is evaluated in zero-shot mode using the `amazon/chronos-2-small` checkpoint with no fine-tuning on Milan data. Quantile forecasts are extracted from sampled token paths. No adaptation is applied.

C. CLASP detector configuration

CLASP parameters are tuned on validation data via a grid search over 36 combinations:

- `min_size` $\in \{8, 10, 12\}$
- `period_length` $\in \{8, 16, \text{auto}\}$
- `score_threshold` $\in \{0.5, 0.6, 0.7, 0.75\}$

The selection criterion is τ-MAE on validation windows where both the predicted and true break fall within the horizon. Break F1 saturates at 1.0 on training data for all configurations and provides no discrimination. The best configuration — min_size $= 8$, period_length $= 16$, score_threshold $= 0.5$ — is shared across all five forecasters.

TABLE III
FORECAST METRICS ON THE TIM MILAN TEST SET (7,400 WINDOWS). LOWER IS BETTER FOR RMSE AND PINBALL. COVERAGE TARGET: 0.95.

Model	RMSE_{q50}	Pinball_{q50}	Pinball_{q95}	Cover_{q95}
TFT	**5.658**	**1.152**	**0.357**	0.924
LSTM	5.684	1.389	0.385	**0.951**
DeepAR	5.796	1.493	0.817	0.667
Seasonal Naive	7.263	1.405	0.489	0.858
Chronos-2	7.071	2.108	0.728	0.814

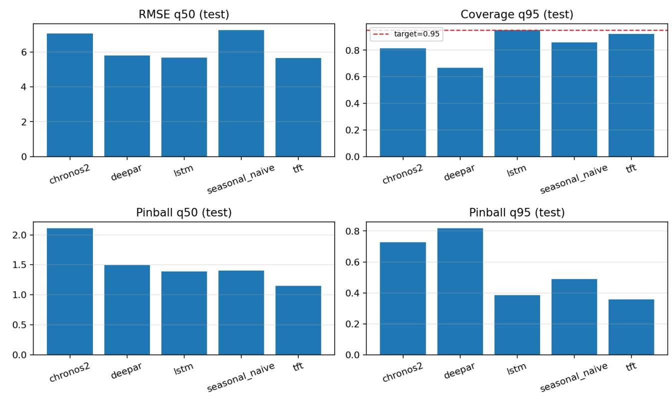

Fig. 2. Forecast metrics on the TIM Milan test set. Left: RMSE and pinball at $q_{0.5}$. Right: pinball at $q_{0.95}$ and coverage. The 0.95 target is shown as a dashed line on the coverage panel.

D. Evaluation metrics

Forecast quality: RMSE on the median path, pinball loss at $q \in \{0.5, 0.95\}$, q95 coverage (fraction of realized values below $q_{0.95}$), and interval width.

System quality: change-point timing MAE (MAE_{CP}), tolerance hit rate (fraction of windows with $|\tau_{\text{pred}} - \tau_{\text{true}}| \leq 3$ steps), pre-break coverage rate (fraction of realized pre-break values below c_{safe}), and sharpness ($c_{\text{safe}} - \max(\mathcal{I}_{\text{true}})$). The tolerance threshold of 3 steps corresponds to 30 minutes at the native 10-minute cadence.

VIII. RESULTS

A. Forecast quality

Table III and Figure 2 show forecast results across all five models. TFT leads on point accuracy (RMSE 5.658, pinball_{q50} 1.152) and upper-path precision (pinball_{q95} 0.357). LSTM is close on RMSE (5.684) and achieves the strongest coverage (0.951, meeting the 0.95 target). Seasonal Naive is worst on RMSE (7.263) but reaches reasonable coverage (0.858).

The two failure modes appear in the remaining models. DeepAR achieves competitive RMSE (5.796) but its Gaussian intervals are too narrow: coverage 0.667, more than 28 points below target. Chronos-2 falls short on RMSE (7.071) and coverage (0.814), expected for a zero-shot model on this domain, but produces sharper upper-path estimates than DeepAR (pinball_{q95} 0.728 vs. 0.817).

TABLE IV
SYSTEM METRICS ON THE TIM MILAN TEST SET. REL. SHARP. $= (c_{\text{safe}} - \text{PEAK})/\text{PEAK}$; LOWER IS BETTER. COVERAGE TARGET: ≥ 0.95.

Model	MAE_{CP}	Tol. hit	Coverage	Rel. Sharp.
LSTM	**7.175**	**0.487**	**0.984**	0.237
TFT	7.386	0.462	0.968	0.153
Seasonal Naive	7.333	0.427	0.973	0.251
Chronos-2	7.415	0.420	0.873	**0.082**
DeepAR	7.162	0.427	0.802	−0.041

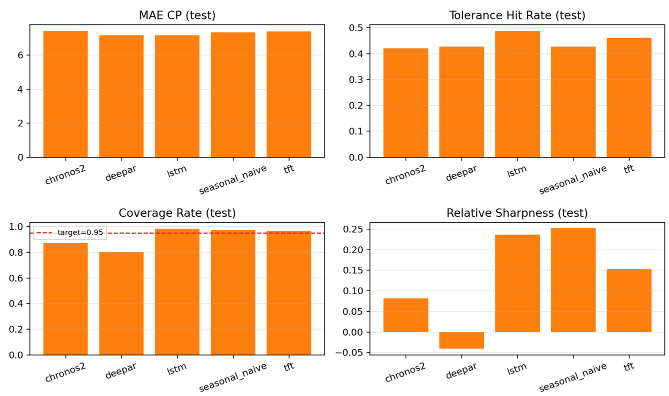

Fig. 3. System metrics after break-point detection and safe-ceiling construction. The coverage panel shows the 0.95 target as a dashed line. DeepAR is the only model with negative sharpness (ceiling below realized traffic peak).

B. System quality

Table IV and Figure 3 show the system results. Three findings dominate.

Coverage vs. relative sharpness tradeoff. No model dominates both metrics. LSTM maximizes coverage (0.984) with the widest ceiling (relative sharpness 0.237: ceiling is 23.7% above the realized demand peak). Chronos-2 achieves the tightest ceiling in the benchmark (relative sharpness 0.082: only 8.2% above peak) — 65% tighter than LSTM relative to demand — at the cost of coverage (0.873). TFT occupies the best trained operating point: coverage 0.968 (above the 0.95 target), relative sharpness 0.153 (35% tighter than LSTM). Seasonal Naive is safe but wasteful: coverage 0.973, relative sharpness 0.251 (25.1% excess). The tradeoff is real and operators must choose a point on it based on their provisioning risk tolerance.

CP timing. All models fall within 0.25 steps of each other (MAE_{CP} range 7.16–7.42). Chronos-2 zero-shot is within 0.24 steps of LSTM. Tolerance hit rates range from 0.420 to 0.487: roughly half of all predicted breaks land within 30 minutes of the true break across all models.

DeepAR failure. Relative sharpness -0.041 means the safe ceiling falls 4.1% below the realized demand peak — the pipeline under-provisions the network. This is not a conservative failure; it is the opposite. The cause is Gaussian confidence intervals that are too narrow for the heavy-tailed traffic distribution, confirmed by the 0.667 q95 forecast coverage. For safety-critical provisioning, negative relative sharpness is disqualifying.

C. Chronos-2 as the LLM6G operating point

Figure 4 illustrates the coverage-sharpness tradeoff on individual windows. TFT sets a more conservative ceiling; Chronos-2 sets a tighter one closer to the realized traffic. The Chronos-2 window was produced with no Milan-specific training.

Chronos-2 achieves the tightest ceiling (relative sharpness 0.082: 8.2% excess above demand) with competitive CP timing (7.415 MAE_{CP}), 0.24 steps worse than LSTM. The coverage tradeoff (0.873 vs. 0.984 for LSTM) is the explicit cost. In deployment scenarios where traffic distribution shifts make periodic retraining expensive, Chronos-2 provides an immediately deployable operating point with documented tradeoffs.

D. Post-hoc τ calibration (negative result)

Post-hoc calibration of τ_{pred} was evaluated using random forests, gradient boosting, and per-series offset corrections on validation windows. All calibrators produce zero improvement on test data. After the CLASP validation sweep, the detector is already at near-optimal performance — no residual calibration signal exists. This negative result is informative: it confirms that the sweep-selected CLASP configuration is sufficient and further post-processing does not help.

IX. DISCUSSION

A. Pipeline reusability and the coverage-sharpness tradeoff

The five-model benchmark demonstrates that the pipeline accepts any forecaster that produces $(q_{0.5}, q_{0.95})$ over a horizon. Different forecasters produce different points on the coverage vs. relative sharpness tradeoff curve — and this tradeoff is a property of the forecaster and the traffic distribution, not an artifact of evaluation. Operators with strict SLA requirements (coverage > 0.95) should prefer LSTM or TFT. Operators optimizing for resource efficiency who can tolerate a coverage margin reduction should prefer Chronos-2.

B. Chronos-2 and the LLM6G vision

The central result for the LLM6G framing is that Chronos-2 achieves the tightest safe ceiling (relative sharpness 0.082: 8.2% excess above realized demand) across all five models without any domain training. Under recurring distribution shifts, operator-grade ML deployment requires retraining cycles whenever the traffic regime changes significantly. Foundation models sidestep this: Chronos-2 enters the control loop immediately with a documented tradeoff (coverage 0.873 vs. LSTM's 0.984). Whether that tradeoff is acceptable depends on deployment risk tolerance.

The CP timing gap confirms that Chronos-2 is competitive at regime boundary detection: 7.415 MAE_{CP} vs. 7.175 for LSTM, a difference of 0.24 steps (2.4 minutes). Roughly 42% of Chronos-2 break predictions land within 30 minutes of the true break.

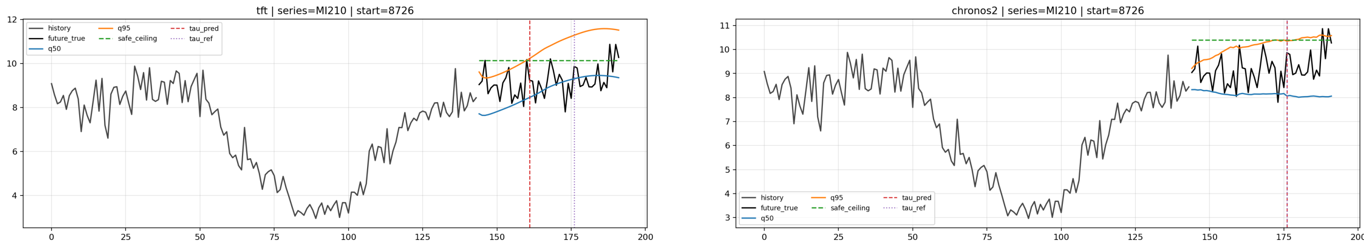

Fig. 4. Example system windows on the test set. Left: TFT (coverage 0.968, rel. sharpness 0.153). Right: Chronos-2 (coverage 0.873, rel. sharpness 0.082). Both show history, forecast quantiles, predicted break, true break, and safe ceiling. The Chronos-2 ceiling sits closer to the realized traffic peak.

C. TFT and channel-selective normalization

TFT is the best trained model in the benchmark, enabled by a non-trivial implementation decision. Multi-channel input normalization must be applied selectively: raw traffic on channel 0 is normalized by instance mean, but time-feature channels encoding cyclic calendar position must not be rescaled. Normalizing all channels erases the calendar signal — the validation-to-test RMSE gap exceeded 30% under incorrect normalization. With channel-selective normalization, TFT achieves RMSE 5.658 and relative sharpness 0.153.

This is an example of the broader lesson: implementation decisions at the data preprocessing boundary determine whether a trained model generalizes to new distribution regimes.

D. DeepAR as a cautionary case

DeepAR's relative sharpness of -0.041 (ceiling 4.1% below realized demand peak) makes it a documented failure case for this pipeline. The model architecture is sound; the failure comes from likelihood mismatch. Milan traffic values are continuous grid-cell aggregates, not integer counts, yet the Gaussian assumption still produces overconfident intervals. The practical lesson: interval calibration must be verified on the control metric (sharpness, coverage), not just on the distributional fit metric (NLL). When NLL was used as the training monitor, NLL improved while coverage stagnated at 0.64 — the best-NLL checkpoint was not the best quantile checkpoint.

E. Context length and stationarity

The 144-step (24-hour) context captures the full daily cycle, which is the dominant periodicity in urban telecom traffic. Shorter contexts miss this cycle. Longer contexts risk including multiple regime changes in the conditioning window, which degrades instance normalization and forces the model to handle multi-regime inputs. The 144-step context is appropriate for this data.

F. Limitations and future work

The benchmark uses 200 cells, a neutral ascending-ID subset of the full 8,883-cell TIM Milan dataset. All claims are at the level of public grid-cell telecom proxies, not operator base-station telemetry. The evaluation is offline; no real-time receding-horizon controller is implemented.

Three natural extensions follow from this work. First, scaling to the full 8,883-cell benchmark tests whether the findings hold at full city scale. Second, evaluation on operator-grade sector telemetry tests the transfer from public proxy to production data. Third, coverage-constrained fine-tuning of Chronos-2 would test whether the foundation model can close the coverage gap ($0.873 \to 0.95$) while retaining its sharpness advantage — a direct operationalization of the LLM6G vision.

X. Conclusion

We present LLM6G: a shift-aware forecast-to-control pipeline for mobile network traffic, benchmarked on the public TIM Milan dataset with five forecasters. The pipeline makes the next distribution break a first-class control object — converting any probabilistic forecast into a safe operating ceiling over the predicted stationary interval and refreshing after the break. Any forecaster that produces $(q_{0.5}, q_{0.95})$ over a horizon plugs into the same interface.

Coverage and relative sharpness compete. TFT achieves the best trained operating point (coverage 0.968, relative sharpness 0.153: 15.3% excess above demand). Chronos-2, with no domain training, achieves the tightest ceiling in the benchmark (relative sharpness 0.082: 8.2% excess) — demonstrating that LLM-based foundation models can operate directly in 6G-oriented control loops. DeepAR relative sharpness of -0.041 documents what happens when likelihood mismatch propagates through the pipeline: the ceiling falls below realized demand.

Three implementation decisions determined whether the loop works or fails silently: channel-selective normalization for TFT, Gaussian likelihood for continuous traffic values, and quantile-based checkpoint monitoring instead of distributional fit. These decisions are documented as reproducible contributions.

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